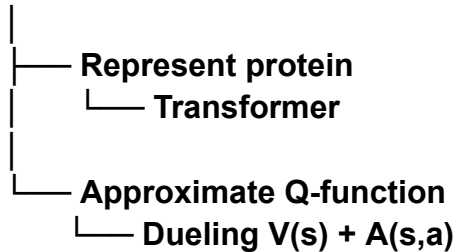


The basic idea is to build the protein one residue at a time, let an RL agent decide where each residue should go, and reward the agent when hydrophobic residues form more contacts.

The agent learns this through:

DuelingTransformerQNetwork



The project uses a Dueling DQN whose Q-network is based on a Transformer.

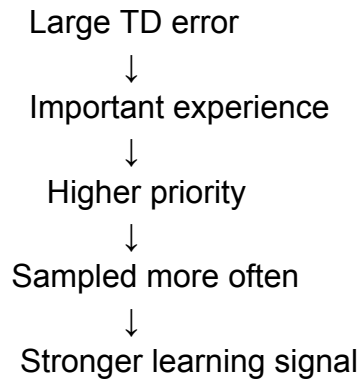
DQN [This learns the Q-Function, given the state what should be the action for max reward]
+
Transformer [The DQN needs to understand the **state**, ie. models residues and relationship between them]
+
Dueling Architecture [Dueling is a type/variant of DQN - that outputs **V(s)** = how good the current state is
A(s,a) = how advantageous each action is in that state]
+
Prioritized Experience Replay [Prioritized Experience Replay (PER) is about **deciding** which past experiences the DQN should learn from first.]

The agent stores experiences: [(s, a, r, s', done)]

Where:

- (s): current state
- (a): action taken
- (r): reward
- (s'): next state
- **done**: episode termination

Experiences with higher TD error are sampled more often.

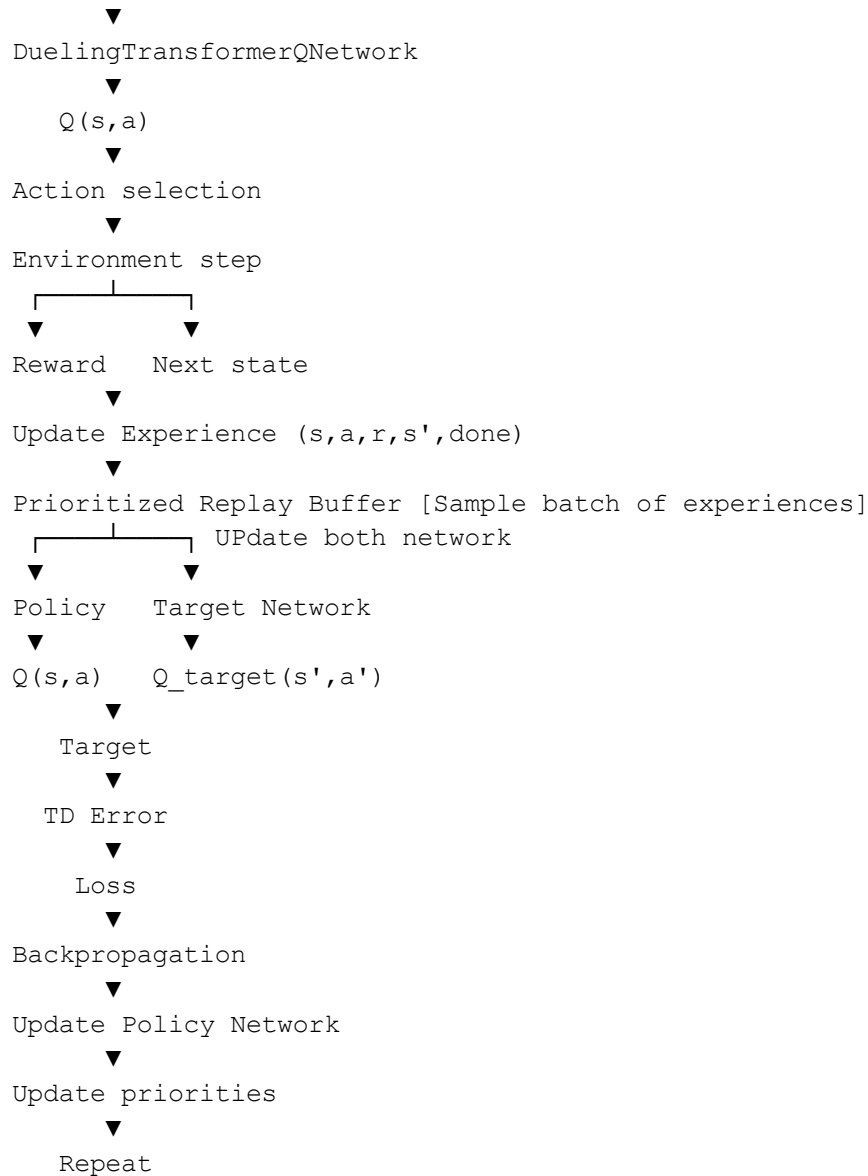


The final objective is to discover a **high-quality 3D protein conformation with a large number of H-H contacts**.

The model uses two identical networks: Policy Network & Target Network:-

- Both networks have the same architecture, but they are used differently.
- Policy network = you are the student who is constantly learning and changing their answers.
- Target network = an answer key that stays fixed for a while.
-
- The policy network is trained continuously:
 - `loss.backward()`
 - `optimizer.step()`
- The target network is NOT directly trained.
- Instead, every 1000 steps, you copy the policy network's weights into it:
 - `self.target_net.load_state_dict(`
 - `self.policy_net.state_dict()`
 - `)`
 -
- The two networks are structurally the same, but the policy network learns, while the target network provides a stable reference for learning.
- Both are learning but at different rates. The policy network is updated every step, while the target network is updated less frequently. This helps stabilize learning by providing a more consistent target for the policy network to learn from.

CURRENT PROTEIN [CURRENT STRUCTURE Si]



Policy Network gives: $Q_{\text{policy}}(s,a)$

Target Network gives future Q-values: $Q_{\text{target}}(s',a')$

But we don't directly compare these two.
Instead, we first create the target:

$\text{TargetQ} = \text{reward} [\text{number of H-H bonds}] + \gamma a' \max Q_{\text{target}}(s',a')$

Then: $\text{TD Error} = \text{TargetQ} - Q_{\text{policy}}(s,a)$

